

Using Machine Learning to Analyze Traffic Flows in Lima, Peru

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As the global population grows exponentially, an increasing proportion of the world has begun to live in urban environments, with an estimated [%] living in cities by []. Accurate traffic flow prediction models are necessary in order to optimize traffic patterns and inform sustainable, safe, and cost-effective infrastructure. Recent literature utilizes advanced machine learning (ML) and deep learning techniques in order to build Intelligent Transportation Systems (ITS). However, the majority of studies concentrate on highly-developed cities where data is readily available, while fewer studies focus on the implementation of ML-powered traffic control monitoring systems in developing cities. Since developing cities are likely to suffer from uneven development which begets traffic congestion and traffic-related accidents, understanding traffic flows in developing cities is extremely necessary for effective urban development.

XGBoost is found to be more computationally efficient and more accurate than deep learning methods on sparse data. Good at calculating tabular data.

The project will utilize Open-Source Map (OSM), satellite imagery (basic feature)

In your proposal, you will provide justification on why your project matters based on work which has been done in this area, using in-line citations [?] to refer to existing works. Include what your project will accomplish and how your software will function.

When you use an image, such as in Figure 1, refer to it in the text.



Figure 1: Archie

Appendix

A concise list of features / user stories in the order in which they will be built. A few examples are below to demonstrate the expected scope and level of granularity; you will have more features than this.

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- Statistical baseline (historical average / ARIMA)
- XGBoost based on
- Feature importance
- SHAP interpretation
- Cross validation with MAE or RMSE
- Scenario testing
- Use GIS software to make a dynamic model that can map results

Problem definition (city + data)

Data gathering and cleaning

Baseline models - graph based models (graph neural networks, graph + lstm/gru) - traffic grid (cnn, lstm for spatial convolutions) - bayesian models for handling missing data

Spatiotemporal model

Evaluation (MAE, RMSE, MAPE) Evaluation of efficiency (compare to known values / bayesian uncertainty)

Interpretation and insights

Policy or planning implications

References